

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

1	Monday 1 June 2026	2
1.1	Daily Limit	2
2	Tuesday 2 June 2026	3
2.1	Daily Limit	3
3	Wednesday 3 June 2026	4
3.1	Daily Limit	4
4	Thursday 4 June 2026	5
4.1	Daily Limit	5
5	Friday 5 June 2026	7
5.1	Daily Limit	7
6	Saturday 6 June 2026	8
6.1	Daily Limit	8
7	Sunday 7 June 2026	9
7.1	Daily Limit	9
8	Monday 8 June 2026	10
8.1	Daily Limit	10
9	Tuesday 9 June 2026	11
9.1	Daily Limit	11
10	Wednesday 10 June 2026	12
10.1	Daily Limit	12

Monday 1 June 2026

1.1 Daily Limit

Limit involving the Lambert W Function

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{(W_0(x) - x + x^2)(\sin(x) - x)}{x^6}$$

Solution: To evaluate this limit, we use the Taylor series expansions of the functions around $x = 0$. The expansion for the principal branch of the Lambert W function, $W_0(x)$, is:

$$W_0(x) = x - x^2 + \frac{3}{2}x^3 - \frac{8}{3}x^4 + O(x^5)$$

Substituting this into the first term of the numerator gives:

$$W_0(x) - x + x^2 = \left(x - x^2 + \frac{3}{2}x^3 - \dots\right) - x + x^2 = \frac{3}{2}x^3 + O(x^4)$$

The Maclaurin series for the sine function is:

$$\sin(x) = x - \frac{x^3}{6} + \frac{x^5}{120} - \dots$$

Substituting this into the second term of the numerator gives:

$$\sin(x) - x = \left(x - \frac{x^3}{6} + \dots\right) - x = -\frac{x^3}{6} + O(x^5)$$

Now, multiply the two leading terms in the numerator:

$$\left(\frac{3}{2}x^3\right) \left(-\frac{x^3}{6}\right) = -\frac{3}{12}x^6 = -\frac{1}{4}x^6$$

The overall limit expression becomes:

$$\lim_{x \rightarrow 0} \frac{-\frac{1}{4}x^6 + O(x^7)}{x^6} = -\frac{1}{4}$$

Tuesday 2 June 2026

2.1 Daily Limit**Limit of an Exponential and Factorial Function**

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\sqrt{e^x} - x! \tan(x - \sin(x))}{x}$$

Solution: Let us evaluate the behavior of the numerator and the denominator as $x \rightarrow 0$.

For the numerator, we look at the individual terms: 1. As $x \rightarrow 0$, $\sqrt{e^x} = e^{x/2} \rightarrow e^0 = 1$. 2. The term $x!$ (which formally is the Gamma function $\Gamma(x + 1)$ for real numbers) approaches $0! = 1$ as $x \rightarrow 0$. 3. The term $\tan(x - \sin(x))$ approaches $\tan(0 - 0) = 0$.

Thus, the second part of the numerator $x! \tan(x - \sin(x))$ approaches $1 \cdot 0 = 0$. The entire numerator approaches:

$$1 - 0 = 1$$

The denominator approaches 0. Since the numerator approaches a non-zero constant (1) while the denominator approaches 0, the overall fraction grows without bound. Specifically, as $x \rightarrow 0^+$, the limit approaches $+\infty$, and as $x \rightarrow 0^-$, it approaches $-\infty$.

Therefore, the two-sided limit does not exist (or approaches ∞).

Wednesday 3 June 2026

3.1 Daily Limit

Largest Non-Negative Integer for a Limit

The largest value of the non-negative integer a for which

$$\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$$

is to be determined.

Solution: First, let's simplify the exponent. As $x \rightarrow 1$:

$$\frac{1-x}{1-\sqrt{x}} = \frac{(1-\sqrt{x})(1+\sqrt{x})}{1-\sqrt{x}} = 1+\sqrt{x}$$

Taking the limit as $x \rightarrow 1$, the exponent approaches 2.

Next, let $t = x - 1$. As $x \rightarrow 1$, $t \rightarrow 0$. We can rewrite the base of the limit in terms of t :

$$\frac{-a(t+1) + \sin(t) + a}{(t+1) + \sin(t) - 1} = \frac{-at - a + \sin(t) + a}{t + \sin(t)} = \frac{\sin(t) - at}{\sin(t) + t}$$

Using the Maclaurin expansion $\sin(t) \approx t - \frac{t^3}{6}$, we find the leading terms for the numerator and denominator:

$$\lim_{t \rightarrow 0} \frac{t - at}{t + t} = \lim_{t \rightarrow 0} \frac{t(1-a)}{2t} = \frac{1-a}{2}$$

Putting it all together, the original limit is:

$$\left(\frac{1-a}{2} \right)^2 = \frac{1}{4}$$

Taking the square root of both sides gives:

$$\frac{1-a}{2} = \pm \frac{1}{2} \implies 1-a = \pm 1$$

Solving for a , we get two possible solutions: 1. $1-a = 1 \implies a = 0$ 2. $1-a = -1 \implies a = 2$

Since we are looking for the largest non-negative integer, the answer is:

Thursday 4 June 2026

4.1 Daily Limit

Limit of a Complex Trigonometric Ratio

Evaluate the limit:

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{4\sqrt{2}(\sin(3x) + \sin(x))}{(2\sin(2x)\sin(\frac{3x}{2}) + \cos(\frac{5x}{2})) - (\sqrt{2} + \sqrt{2}\cos(2x) + \cos(\frac{3x}{2}))}$$

Solution: Let $x = \frac{\pi}{2} + t$. As $x \rightarrow \frac{\pi}{2}$, we have $t \rightarrow 0$. We apply this substitution to simplify the trigonometric terms in the numerator. Using sum-to-product formulas, the numerator is:

$$4\sqrt{2}(\sin(3x) + \sin(x)) = 4\sqrt{2}(2\sin(2x)\cos(x)) = 8\sqrt{2}\sin(2x)\cos(x)$$

Substituting $x = \frac{\pi}{2} + t$:

$$\sin(2(\frac{\pi}{2} + t)) = \sin(\pi + 2t) = -\sin(2t)$$

$$\cos(\frac{\pi}{2} + t) = -\sin(t)$$

So the numerator becomes:

$$8\sqrt{2}(-\sin(2t))(-\sin(t)) = 8\sqrt{2}\sin(2t)\sin(t) \approx 8\sqrt{2}(2t)(t) = 16\sqrt{2}t^2$$

For the denominator $D(x)$, we can simplify the product of sines using the product-to-sum formula $2\sin(2x)\sin(\frac{3x}{2}) = \cos(\frac{x}{2}) - \cos(\frac{7x}{2})$:

$$D(x) = \cos(\frac{x}{2}) - \cos(\frac{7x}{2}) + \cos(\frac{5x}{2}) - \sqrt{2} - \sqrt{2}\cos(2x) - \cos(\frac{3x}{2})$$

We evaluate the Maclaurin series of $D(t)$ around $t = 0$ by finding the derivatives of $D(x)$ at $x = \frac{\pi}{2}$. Evaluating $D(\frac{\pi}{2})$ yields:

$$\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} - \sqrt{2} - \sqrt{2}(-1) - \left(-\frac{\sqrt{2}}{2}\right) = 0$$

The first derivative $D'(x)$ is:

$$-\frac{1}{2}\sin(\frac{x}{2}) + \frac{7}{2}\sin(\frac{7x}{2}) - \frac{5}{2}\sin(\frac{5x}{2}) + 2\sqrt{2}\sin(2x) + \frac{3}{2}\sin(\frac{3x}{2})$$

Evaluating $D'(\frac{\pi}{2})$ gives:

$$\frac{\sqrt{2}}{4}(-1 - 7 + 5 + 3) + 0 = 0$$

The second derivative $D''(x)$ is:

$$-\frac{1}{4}\cos(\frac{x}{2}) + \frac{49}{4}\cos(\frac{7x}{2}) - \frac{25}{4}\cos(\frac{5x}{2}) + 4\sqrt{2}\cos(2x) + \frac{9}{4}\cos(\frac{3x}{2})$$

Evaluating $D''(\frac{\pi}{2})$ gives:

$$\frac{\sqrt{2}}{8}(-1 + 49 + 25 - 9) - 4\sqrt{2} = 8\sqrt{2} - 4\sqrt{2} = 4\sqrt{2}$$

Thus, the Taylor expansion of the denominator is $D(t) \approx \frac{D''(\pi/2)}{2}t^2 = 2\sqrt{2}t^2$.

Evaluating the ratio of the leading terms gives:

$$\lim_{t \rightarrow 0} \frac{16\sqrt{2}t^2}{2\sqrt{2}t^2} = 8$$

Friday 5 June 2026

5.1 Daily Limit

Limit of an Exponential Sum Ratio

Evaluate the limit:

$$\lim_{n \rightarrow \infty} \frac{1^n + 2^n + \cdots + n^n}{n + n^2 + \cdots + n^n}$$

Solution: Let's analyze the dominant terms of the sums in the numerator and denominator as $n \rightarrow \infty$.

For the denominator, we have a geometric series:

$$\sum_{k=1}^n n^k = n \frac{n^n - 1}{n - 1}$$

As $n \rightarrow \infty$, this sum is strictly dominated by its largest term, yielding approximately n^n . More precisely, dividing by n^n :

$$\frac{\sum_{k=1}^n n^k}{n^n} = 1 + \frac{1}{n} + \frac{1}{n^2} + \cdots \rightarrow 1$$

For the numerator, the terms are of the form k^n . We can reverse the index to observe the largest terms by setting $k = n - j$:

$$\sum_{k=1}^n k^n = n^n + (n-1)^n + (n-2)^n + \cdots$$

Dividing the numerator by n^n gives:

$$\frac{\sum_{k=1}^n k^n}{n^n} = 1 + \left(1 - \frac{1}{n}\right)^n + \left(1 - \frac{2}{n}\right)^n + \cdots$$

Recall the standard limit definition of e :

$$\lim_{n \rightarrow \infty} \left(1 - \frac{j}{n}\right)^n = e^{-j}$$

By the dominated convergence theorem, the limit of the sum is the sum of the limits:

$$\lim_{n \rightarrow \infty} \frac{1}{n^n} \sum_{k=1}^n k^n = \sum_{j=0}^{\infty} e^{-j}$$

This is a convergent infinite geometric series with a first term of 1 and a common ratio of $1/e$:

$$\sum_{j=0}^{\infty} (e^{-1})^j = \frac{1}{1 - e^{-1}} = \frac{e}{e - 1}$$

Since the denominator normalized by n^n approaches 1, the original limit evaluates strictly to:

$$\frac{e}{e - 1}$$

Saturday 6 June 2026

6.1 Daily Limit

Limit with Binomial and Radical Expansions

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{(1+4x)^5 - (1+5x)^4}{\sqrt[4]{1+4x} - \sqrt[5]{1+5x}}$$

Solution: We utilize the Binomial series expansion $(1+u)^n \approx 1 + nu + \frac{n(n-1)}{2}u^2$ for the terms as $x \rightarrow 0$.

For the numerator:

$$(1+4x)^5 = 1 + 5(4x) + \frac{5 \cdot 4}{2}(4x)^2 + O(x^3) = 1 + 20x + 160x^2 + O(x^3)$$

$$(1+5x)^4 = 1 + 4(5x) + \frac{4 \cdot 3}{2}(5x)^2 + O(x^3) = 1 + 20x + 150x^2 + O(x^3)$$

Subtracting these gives the numerator:

$$(1 + 20x + 160x^2) - (1 + 20x + 150x^2) = 10x^2 + O(x^3)$$

For the denominator, we use the generalized binomial expansion $(1+u)^{1/k}$:

$$\sqrt[4]{1+4x} = (1+4x)^{1/4} = 1 + \frac{1}{4}(4x) + \frac{\frac{1}{4}(\frac{1}{4}-1)}{2}(4x)^2 + O(x^3)$$

$$= 1 + x + \frac{-\frac{3}{16}}{2}(16x^2) = 1 + x - \frac{3}{2}x^2 + O(x^3)$$

$$\sqrt[5]{1+5x} = (1+5x)^{1/5} = 1 + \frac{1}{5}(5x) + \frac{\frac{1}{5}(\frac{1}{5}-1)}{2}(5x)^2 + O(x^3)$$

$$= 1 + x + \frac{-\frac{4}{25}}{2}(25x^2) = 1 + x - 2x^2 + O(x^3)$$

Subtracting these gives the denominator:

$$\left(1 + x - \frac{3}{2}x^2\right) - (1 + x - 2x^2) = \left(-\frac{3}{2} - (-2)\right)x^2 = \frac{1}{2}x^2 + O(x^3)$$

Evaluating the limit by taking the ratio of the leading terms:

$$\lim_{x \rightarrow 0} \frac{10x^2}{\frac{1}{2}x^2} = 20$$

Sunday 7 June 2026

7.1 Daily Limit

Limit Involving Logarithms and Inverse Hyperbolic Functions

Evaluate the limit:

$$\lim_{x \rightarrow 0} \left(\frac{1}{\ln(x + \sqrt{x^2 + 1})} - \frac{1}{\ln(1 + x)} \right)$$

Solution: Recognize that $\ln(x + \sqrt{x^2 + 1})$ is the definition of the inverse hyperbolic sine function, $\operatorname{arsinh}(x)$. We find the Maclaurin series for both logarithmic terms. For the first term:

$$\operatorname{arsinh}(x) = x - \frac{x^3}{6} + O(x^5)$$

For the second term:

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - O(x^4)$$

Substitute these into the original expression:

$$\frac{1}{x - \frac{x^3}{6}} - \frac{1}{x - \frac{x^2}{2} + \frac{x^3}{3}} = \frac{1}{x} \left(\frac{1}{1 - \frac{x^2}{6}} - \frac{1}{1 - \frac{x}{2} + \frac{x^2}{3}} \right)$$

Using the geometric series expansion $\frac{1}{1-u} \approx 1 + u + u^2$:

$$\frac{1}{1 - \frac{x^2}{6}} = 1 + \frac{x^2}{6} + O(x^4)$$

For the second fraction, let $u = \frac{x}{2} - \frac{x^2}{3}$:

$$\begin{aligned} \frac{1}{1 - \left(\frac{x}{2} - \frac{x^2}{3}\right)} &= 1 + \left(\frac{x}{2} - \frac{x^2}{3}\right) + \left(\frac{x}{2} - \frac{x^2}{3}\right)^2 + O(x^3) \\ &= 1 + \frac{x}{2} - \frac{x^2}{3} + \frac{x^2}{4} + O(x^3) = 1 + \frac{x}{2} - \frac{x^2}{12} + O(x^3) \end{aligned}$$

Subtracting the expanded forms inside the parenthesis:

$$\left(1 + \frac{x^2}{6}\right) - \left(1 + \frac{x}{2} - \frac{x^2}{12}\right) = -\frac{x}{2} + \left(\frac{1}{6} + \frac{1}{12}\right)x^2 = -\frac{x}{2} + \frac{x^2}{4} + O(x^3)$$

Multiplying by $\frac{1}{x}$ from outside:

$$\frac{1}{x} \left(-\frac{x}{2} + \frac{x^2}{4} \right) = -\frac{1}{2} + \frac{x}{4}$$

Taking the limit as $x \rightarrow 0$, the remaining x terms vanish, leaving:

$$-\frac{1}{2}$$

Monday 8 June 2026

8.1 Daily Limit

Limit of Nested Trigonometric Functions

Evaluate the limit:

$$\lim_{x \rightarrow 0} \frac{\cos(\sin(x)) - \cos(x)}{x^4}$$

Solution: We use the Maclaurin series for both the inner and outer trigonometric functions. First, the series for $\cos(x)$:

$$\cos(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - O(x^6)$$

Next, we need the series for $\cos(\sin(x))$. Since $\sin(x) = x - \frac{x^3}{6} + O(x^5)$, we substitute this into the cosine expansion:

$$\cos(\sin(x)) = 1 - \frac{1}{2}(\sin(x))^2 + \frac{1}{24}(\sin(x))^4 - \dots$$

Calculate the powers of $\sin(x)$:

$$(\sin(x))^2 = \left(x - \frac{x^3}{6}\right)^2 = x^2 - 2(x)\left(\frac{x^3}{6}\right) + O(x^6) = x^2 - \frac{x^4}{3} + O(x^6)$$

$$(\sin(x))^4 = x^4 + O(x^6)$$

Substitute these back into the expression for $\cos(\sin(x))$:

$$\begin{aligned} \cos(\sin(x)) &= 1 - \frac{1}{2}\left(x^2 - \frac{x^4}{3}\right) + \frac{1}{24}(x^4) + O(x^6) \\ &= 1 - \frac{x^2}{2} + \frac{x^4}{6} + \frac{x^4}{24} = 1 - \frac{x^2}{2} + \left(\frac{4}{24} + \frac{1}{24}\right)x^4 = 1 - \frac{x^2}{2} + \frac{5x^4}{24} + O(x^6) \end{aligned}$$

Now we find the numerator by subtracting $\cos(x)$:

$$\begin{aligned} \cos(\sin(x)) - \cos(x) &= \left(1 - \frac{x^2}{2} + \frac{5x^4}{24}\right) - \left(1 - \frac{x^2}{2} + \frac{x^4}{24}\right) \\ &= \frac{5x^4}{24} - \frac{x^4}{24} = \frac{4x^4}{24} = \frac{x^4}{6} \end{aligned}$$

Dividing by x^4 and evaluating the limit:

$$\lim_{x \rightarrow 0} \frac{\frac{x^4}{6}}{x^4} = \frac{1}{6}$$

Tuesday 9 June 2026

9.1 Daily Limit**Limit of an Exponential Form as $x \rightarrow \infty$**

Evaluate the limit:

$$\lim_{x \rightarrow +\infty} (xe^x)^{1/x}$$

Solution: We can simplify the expression using exponent rules:

$$\begin{aligned}(xe^x)^{1/x} &= x^{1/x} \cdot (e^x)^{1/x} \\ &= x^{1/x} \cdot e^{x/x} = x^{1/x} \cdot e^1 = e \cdot x^{1/x}\end{aligned}$$

Now, evaluate the limit of the remaining x -dependent factor as $x \rightarrow \infty$. We know that:

$$\lim_{x \rightarrow \infty} x^{1/x}$$

Let $y = x^{1/x}$, so $\ln(y) = \frac{\ln(x)}{x}$. Applying L'Hôpital's rule to the limit of $\ln(y)$:

$$\lim_{x \rightarrow \infty} \frac{\ln(x)}{x} = \lim_{x \rightarrow \infty} \frac{1/x}{1} = 0$$

Since $\ln(y) \rightarrow 0$, $y \rightarrow e^0 = 1$. Therefore, $x^{1/x} \rightarrow 1$.

Substitute this back into our simplified expression:

$$\lim_{x \rightarrow +\infty} (e \cdot x^{1/x}) = e \cdot 1 = e$$

Wednesday 10 June 2026

10.1 Daily Limit

Limit of an Exponential with Variables in Base and Power

Evaluate the limit:

$$\lim_{x \rightarrow \infty} x^{-x + \frac{1}{x}}$$

Solution: Let's analyze the behavior of the base and the exponent independently as $x \rightarrow \infty$. The base is simply x , which approaches ∞ . The exponent is $-x + \frac{1}{x}$. As $x \rightarrow \infty$, the term $\frac{1}{x} \rightarrow 0$, so the exponent approaches $-\infty$.

This results in the form $\infty^{-\infty}$. We can rewrite this algebraically to make it clearer:

$$x^{-x + \frac{1}{x}} = \frac{1}{x^{x - \frac{1}{x}}}$$

As $x \rightarrow \infty$, $x - \frac{1}{x} \rightarrow \infty$. Thus, we are evaluating:

$$\frac{1}{\infty^\infty} = \frac{1}{\infty}$$

Therefore, the limit evaluates directly to:

$$0$$